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Semiconductor package and method of fabricating the same

Abstract

Disclosed are semiconductor packages and their fabrication methods. The semiconductor package comprises semiconductor chips stacked on a substrate and including first and second pads on top surfaces thereof, and bonding wires connecting the first and second pads to the substrate. The semiconductor chips alternately protrude in a first direction and its opposite direction. The semiconductor chip has a first lateral surface spaced apart from another semiconductor chip. The top surface of the semiconductor chip is provided thereon with a first arrangement line extending along the first lateral surface and with second arrangement lines extending from opposite ends of the first arrangement line. Wherein as a distance between the first and second arrangement lines increases, a distance between the second arrangement lines and the first lateral surface increases. The first pads are arranged along the first arrangement line. The second pads are arranged along the second arrangement lines.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This U.S. nonprovisional application claims priority under 35 U.S.C § 119 to Korean Patent Application No. 10-2022-0042479 filed on Apr. 5, 2022 in the Korean Intellectual Property Office, the disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND

(2) The present example inventive concepts relate to a semiconductor package and a method of fabricating the same, and more particularly, to a semiconductor package including semiconductor chips and a method of fabricating the same.

(3) A typical stack package has a structure in which a plurality of devices are stacked. For example, the stack package may include semiconductor chips that are sequentially stacked on a printed circuit board (PCB). Connection pads are formed on the semiconductor chips. The connection pads may be connected through a bonding wire to allow the semiconductor chips to electrically connect to the printed circuit board. The printed circuit board is provided thereon with a logic chip that controls the semiconductor chips.

(4) Portable devices have been increasingly demanded in recent electronic product markets, and as a result, it has been ceaselessly required for reduction in size and weight of electronic parts mounted on the portable devices. In order to accomplish the reduction in size and weight of the electronic parts, there is need for technology to integrate a number of individual devices into a single package as well as technology to reduce individual sizes of mounting parts. In particular, with the integration of a plurality of devices, a semiconductor package requires not only excellent structural properties but also superior electrical properties.

SUMMARY

(5) Some embodiments of the present example inventive concepts provide a semiconductor package with increased structural stability and a method of fabricating the same.

(6) Some embodiments of the present example inventive concepts provide a compact-sized semiconductor package and a method of fabricating the same.

(7) Some embodiments of the present example inventive concepts provide a semiconductor fabrication method with less occurrence of failure and a semiconductor package fabricated by the same.

(8) According to some example embodiments of the present inventive concepts, a semiconductor package may comprise: a substrate; a plurality of semiconductor chips stacked on the substrate and each of the plurality of semiconductor chips including a plurality of first pads and a plurality of second pads on top surfaces of the plurality of semiconductor chips; and a plurality of bonding wires that connect the plurality of first pads and the plurality of second pads to the substrate. The

plurality of semiconductor chips may protrude alternately in a first direction and a direction opposite to the first direction. Protruding distances of the plurality of semiconductor chips may be the same as each other. The plurality of semiconductor chips may have a first lateral surface spaced apart from another semiconductor chip in overlying or underlying contact with the semiconductor chip. The top surface of each of the plurality of semiconductor chips may be provided thereon with a first arrangement line that extends along the first lateral surface and with a plurality of second arrangement lines that extend from opposite ends of the first arrangement line. Wherein as a distance between the second arrangement lines and the first arrangement line increases, a distance between the second arrangement lines and the first lateral surface increases. The plurality of first pads may be arranged along the first arrangement line. The plurality of second pads may be arranged along the second arrangement lines.

(9) According to some example embodiments of the present inventive concepts, a semiconductor package may comprise: a substrate; a plurality of first semiconductor chips and a plurality of second semiconductor chips that are alternately stacked on the substrate, the plurality of second semiconductor chips being shifted in a first direction from the plurality of first semiconductor chips; a plurality of first bonding wires that connect the substrate to the plurality of second semiconductor chips and are bonded to the plurality of second semiconductor chips; and a molding layer on the substrate, the molding layer covering the plurality of first semiconductor chips and the plurality of second semiconductor chips. Each of the plurality of second semiconductor chips may have a first lateral surface positioned in the first direction and a plurality of second lateral surfaces in contact with the first lateral surface. The plurality of first bonding wires may include a plurality of first sub-wires each having a first end portion and a plurality of second sub-wires each having a second end portion. The first end portions may be arranged on a first line that extends along the first lateral surface. The second end portions may be arranged on a plurality of second lines that run across the first lateral surface and one of the second lateral surfaces. On one of the plurality of second semiconductor chips, the first end portions of the first sub-wires may be closer to the first lateral surface than the second end portions of the second sub-wires, and as a distance between the second sub-wires and the second lateral surfaces decreases a distance between the second end portions and the first lateral surface increases. As a distance between the substrate and the plurality of second semiconductor chips increases, a distance between the first lateral surface and the second end portions of the second sub-wires increases.

(10) According to some example embodiments of the present inventive concepts, a method of fabricating a semiconductor package may comprise: placing a first semiconductor chip on a substrate; stacking on the first semiconductor chip a second semiconductor chip that is shifted in a first direction from the first semiconductor chip, the second semiconductor chip including a plurality of chip pads on a top surface of the second semiconductor chip and adjacent to a first lateral surface in the first direction of the second semiconductor chip; and using a bonding machine to bond a plurality of bonding wires to the chip pads. The top surface of the second semiconductor chip may have a plurality of chamfer regions on corners adjacent to the first lateral surface. The chip pads may not be on the chamfer regions.

(11) According to some example embodiments of the present inventive concepts, a semiconductor package may comprise: a substrate; and a plurality of semiconductor chips stacked on the substrate. A top surface of each of the plurality of semiconductor chips may have: a connection region adjacent to a first lateral surface of the plurality of semiconductor chips; and a mounting region adjacent to a second lateral surface of the plurality of semiconductor chips. The second lateral surface may be opposite to the first lateral surface. One of the plurality of semiconductor chips may be attached to the mounting region of another of the plurality of semiconductor chips. The another semiconductor chip may be below the one semiconductor chip. The connection region of the another semiconductor chip may be exposed. The connection region may include: a plurality of chamfer regions adjacent to opposite ends of the first lateral surface; and an intermediate region

between the chamfer regions. Each of the plurality of semiconductor chips may include a plurality of first pads and a plurality of second pads on the intermediate region. The first pads may be arranged along the first lateral surface. The second pads may be arranged along boundaries between the intermediate region and the chamfer regions. The first pads may be closer than the second pads to the first lateral surface.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.
- (2) FIGS. 2 to 5 illustrate plan views showing a semiconductor package according to some example embodiments of the present inventive concepts.
- (3) FIG. 6 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.
- (4) FIGS. 7 to 15 illustrate plan views showing a semiconductor package according to some example embodiments of the present inventive concepts.
- (5) FIGS. 16 and 17 illustrate cross-sectional views showing a semiconductor package according to some example embodiments of the present inventive concepts.
- (6) FIGS. 18 to 20 illustrate plan views showing a semiconductor package according to some example embodiments of the present inventive concepts.
- (7) FIGS. 21A to 25A and 23C illustrate cross-sectional views showing a method of fabricating a semiconductor package according to some example embodiments of the present inventive concepts.
- (8) FIGS. 21B to 25B illustrate plan views showing a method of fabricating a semiconductor package according to some example embodiments of the present inventive concepts.

DETAILED DESCRIPTION OF EMBODIMENTS

- (9) The following will now describe a semiconductor package according to the present example embodiments with reference to the accompanying drawings.
- (10) FIG. 1 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts. FIG. 2 illustrates a plan view showing first and second semiconductor chips of a semiconductor package according to some example embodiments of the present inventive concepts. For convenience of description, FIG. 3 illustrates a plan view of regions on a top surface of the second semiconductor chip depicted in FIG. 2, showing a semiconductor package according to some example embodiments of the present inventive concepts.
- (11) Referring to FIGS. 1 and 2, a package substrate **100** may be provided. The package substrate **100** may be a printed circuit board (PCB) having a signal pattern provided on a top surface thereof. The signal pattern may include first and second substrate pads **102a** and **102b**. The package substrate **100** may have a structure in which at least one dielectric pattern and at least one wiring pattern are alternately stacked. The package substrate **100** may be provided with external terminals **104** on a bottom surface thereof. The external terminals **104** may include solder balls, solder bumps, or solder pads. Based on the type of the external terminals **104**, the package substrate **100** may have a ball grid array (BGA) type, a fine ball-grid array (FBGA) type, or a land grid array (LGA) type.
- (12) In this description, a first direction **D1** and a second direction **D2** may be defined to indicate directions that are parallel to a top surface of the package substrate **100** and intersect each other, and a third direction **D3** may be defined to indicate a direction perpendicular to the top surface of the package substrate **100**.
- (13) A chip stack may be provided on the package substrate **100**. The chip stack may include

semiconductor chips **200** and **300** that are stacked in the third direction **D3** on the package substrate **100**. For convenience of description below, a first semiconductor chip **200** may be defined to denote a semiconductor chip at a lower position of the semiconductor chips **200** and **300** depicted in FIGS. **1** to **4**, and a second semiconductor chip **300** may be defined to denote a semiconductor chip that immediately overlies the first semiconductor chip **200**. However, different naming of the first semiconductor chip **200** and the second semiconductor chip **300** may not mean that the first semiconductor chip **200** and the second semiconductor chip **300** are different chips from each other. The first and second semiconductor chips **200** and **300** may be semiconductor chips having substantially the same configuration and shape as each other, or alternatively may be semiconductor chips different from each other. The first and second semiconductor chips **200** and **300** may be memory chips. For example, the first and second semiconductor chips **200** and **300** may include a dynamic random access memory (DRAM).

(14) The first semiconductor chip **200** may be disposed on the package substrate **100**. The first semiconductor chip **200** may have a first lateral surface **200a** and a second lateral surface **200b** that are opposite to each other in the first direction **DE**. For example, the first lateral surface **200a** may be a side surface positioned in a direction opposite to the first direction **D1** of the first semiconductor chip **200**, and the second lateral surface **200b** may be a side surface positioned in the first direction **D1** of the first semiconductor chip **200**.

(15) A top surface **200u** of the first semiconductor chip **200** may have a mounting region to which is attached the second semiconductor chip **300** which will be discussed below, and may also have a connection region that does not overlap the second semiconductor chip **300**. In this description, on the top surface **200u** of the first semiconductor chip **200**, the mounting region may be a section where a different semiconductor chip is attached to, and the connection region may be a section where first chip pads **220** are provided for electrical connection of the first semiconductor chip **200**. The mounting region and the connection region may be identically applicable to the same components of the second semiconductor chip **300**, or to a mounting region **MR** and a connection region **CR** of the second semiconductor chip **300**. The connection region of the first semiconductor chip **200** may be disposed adjacent to the first lateral surface **200a**. The mounting region of the first semiconductor chip **200** may be disposed adjacent to the second lateral surface **200b**.

(16) The first semiconductor chip **200** may be attached through a first adhesive layer **210** to the package substrate **100**. For example, the first adhesive layer **210** may be provided on a bottom surface of the first semiconductor chip **200**, and may adhere the first semiconductor chip **200** to the top surface of the package substrate **100**. The first semiconductor chip **200** may be attached in a face-up state to the package substrate **100**. The top surface **200u** may be an active surface of the first semiconductor chip **200**. For example, the first chip pads **220** may be provided on the connection region on the top surface **200u** of the first semiconductor chip **200**. The first chip pads **220** may be connected to an integrated circuit of the first semiconductor chip **200**. An arrangement and configuration of the first chip pads **220** will be further described in detail below together with an arrangement and configuration of second chip pads **320**.

(17) The second semiconductor chip **300** may be disposed on the first semiconductor chip **200**. The second semiconductor chip **300** may have a third lateral surface **300a** and a fourth lateral surface **300b** that are opposite to each other in the first direction **D1**. For example, the third lateral surface **300a** may be a side surface positioned in the first direction **D1** of the second semiconductor chip **300**, and the fourth lateral surface **300b** may be a side surface positioned in a direction opposite to the first direction **D1** of the second semiconductor chip **300**.

(18) Referring together to FIGS. **1** to **3**, a top surface of the second semiconductor chip **300** may have a mounting region **MR** and a connection region **CR**. The connection region **CR** of the second semiconductor chip **300** may be disposed adjacent to the third lateral surface **300a**. The mounting region **MR** of the second semiconductor chip **300** may be disposed adjacent to the fourth lateral surface **300b**. On the top surface **300u** of the second semiconductor chip **300**, the connection region

CR may indicate a section where second chip pads **320** are provided for electrical connection of the second semiconductor chip **300**. On the top surface **300u** of the second semiconductor chip **300**, the mounting region MR may be a section where the second chip pads **320** are not provided, or alternatively, different semiconductor chips may be attached on the mounting region MR of the second semiconductor chip **300**.

(19) The second semiconductor chip **300** may be attached through a second adhesive layer **310** to the first semiconductor chip **200**. For example, the second adhesive layer **310** may be provided on a bottom surface of the second semiconductor chip **300**, and may adhere the second semiconductor chip **300** to the mounting region on the top surface **200u** of the first semiconductor chip **200**. The second semiconductor chip **300** may be attached in a face-up state to the first semiconductor chip **200**. The top surface **300u** may be an active surface of the second semiconductor chip **300**. For example, second chip pads **320** may be provided on the connection region CR on the top surface **300u** of the second semiconductor chip **300**. The second chip pads **320** may be connected to an integrated circuit of the second semiconductor chip **300**.

(20) The second semiconductor chip **300** may have the second chip pads **320** provided on the connection region CR. The second chip pads **320** may be disposed adjacent to the third lateral surface **300a** of the second semiconductor chip **300**. In this case, the second chip pads **320** may be provided spaced apart from corners of the second semiconductor chip **300**.

(21) For example, the connection region CR may be disposed adjacent to the third lateral surface **300a** of the second semiconductor chip **300**. The connection region CR may have second regions R2 each of which is adjacent to one of the corners of the second semiconductor chip **300**, and may also have a first region R1 adjacent to the third lateral surface **300a** between the second regions R2. Each of the second regions R2 may be disposed on one of the corners. For example, when viewed in a plan view, the second regions R2 may be chamfer regions defined by the third lateral surface **300a**, fifth lateral surfaces **300c**, and second lines L2. In this description, the chamfer region may be a section provided on a corner where two sides meet each other. The fifth lateral surface **300c** may be side surfaces of the second semiconductor chip **300** that are in contact with the third lateral surface **300a** of the second semiconductor chip **300**, and the second lines L2 may each be an imaginary line that runs across the third lateral surface **300a** and one of the fifth lateral surfaces **300c**. The phrase “run(s) across the third lateral surface **300a** and one of the fifth lateral surfaces **300c**” may mean that, when the second line L2 extends when viewed in a plan view, the second line L2 may intersect both of the third lateral surface **300a** and one of the fifth lateral surfaces **300c**. The first region R1 may be an intermediate region other than the second regions R2 on the connection region CR. For example, the second regions R2 may each have a triangular shape which is disposed on a corresponding corner of the second semiconductor chip **300** and whose width decreases with increasing distance from the third lateral surface **300a**, and the first region R1 may have a shape whose width decreases with increasing distance from the third lateral surface **300a**. For example, a portion of the first region R1 may have a trapezoidal shape.

(22) The second chip pads **320** may be provided on the first region R1 on the top surface **300u** of the second semiconductor chip **300**. For example, the second chip pads **320** may be arranged along a first line L1 and the second lines L2 on the first region RE. For example, the first line L1 and the second lines L2 may be imaginary lines, or arrangement lines, that define positions of the second chip pads **320** on the second semiconductor chip **300**. The first line L1 may extend along the third lateral surface **300a**. The first line L1 may be a straight line parallel to the third lateral surface **300a**. The second lines L2 may extend toward the fifth lateral surfaces **300c** from opposite ends of the first line L1. The second lines L2 may become far away from the third lateral surface **300a** as the second lines L2 are directed toward the fifth lateral surfaces **300c** from opposite ends of the first line L1. As discussed in defining the first region R1 and the second regions R2, the second lines L2 may correspond to boundaries between the first region R1 and the second regions R2. The second lines L2 may each be a straight line that is inclined at an angle of about 45 degrees relative

to the third lateral surface **300a**. As used herein, the term “45 degrees” is intended to be understood to include manufacturing or material tolerances and/or a deviation in magnitude or angle from 45 degrees that is less than or equal to $\pm 10\%$. The present inventive concepts, however, are not limited thereto, and angles between the second lines **L2** and the third lateral surface **300a** may be variously changed if necessary.

(23) On the first region **R1**, the second chip pads **320** may include first unit pads **322** arranged along the first line **L1** and second unit pads **324** arranged along the second lines **L2**. The first unit pads **322** may be disposed closer than the second unit pads **324** to the third lateral surface **300a**. The first unit pads **322** may be arranged in the second direction **D2** so as to be parallel to the third lateral surface **300a**. The closer to the fifth lateral surfaces **300c** the second unit pads **324** are disposed, the farther away from the third lateral surface **300a** the second unit pads **324** are disposed. The second chip pads **320** may not be provided on the second region **R2**.

(24) The first and second semiconductor chips **200** and **300** may be disposed in an offset stack structure. For example, the first and second semiconductor chips **200** and **300** may be stacked obliquely in the first direction **D1**, which may result in an ascending stepwise shape. For more detail, the first semiconductor chip **200** may protrude from the second semiconductor chip **300** in a direction opposite to the first direction **D1**, and the first lateral surface **200a** of the first semiconductor chip **200** may be spaced apart from the second semiconductor chip **300**. The second semiconductor chip **300** may protrude in the first direction **D1** from the first semiconductor chip **200**, and the third lateral surface **300a** of the second semiconductor chip **300** may be spaced apart from the first semiconductor chip **200**. As the first and second semiconductor chips **200** and **300** are stacked in a stepwise shape, there may be exposed a portion of the top surface **200u** of the first semiconductor chip **200**, or the mounting region of the first semiconductor chip **200**.

(25) The first semiconductor chip **200** may have the first chip pads **220** provided on the connection region. An arrangement and shape of the first chip pads **220** may be the same as or similar to an arrangement and shape of the second chip pads **320**. For example, a planar shape of the first semiconductor chip **200** may be symmetric in the first direction **D1** to that of the second semiconductor chip **300**. The expression “symmetric in the first direction **D1**” may include the meaning of both line symmetric in the second direction **D2** orthogonal to the first direction **D1** and point symmetric in the third direction **D3** perpendicular to the first direction **D1**. The first chip pads **220** may be disposed adjacent to the first lateral surface **200a** of the first semiconductor chip **200**. For example, the first chip pads **220** may have third unit pads **222** and fourth unit pads **224**. The third unit pads **222** may be arranged in the second direction **D2** so as to be parallel to the first lateral surface **200a**. The closer the fourth unit pads **224** are to one lateral surface in contact with the first lateral surface **200a**, the farther the fourth unit pads **224** are away from the first lateral surface **200a**. The fourth unit pads **224** may be disposed closer than the third unit pads **222** to the one lateral surface. The third unit pads **222** may be disposed closer than the fourth unit pads **224** to the first lateral surface **200a**. The first chip pads **220** may not be provided on a chamfer region adjacent to the corners of the first semiconductor chip **200**. The chamfer region of the first semiconductor chip **200** may correspond to the second region **R2** of the second semiconductor chip **300**.

(26) The first and second semiconductor chips **200** and **300** may be wire-bonded to the package substrate **100**. The first and second semiconductor chips **200** and **300** may be connected through first and second bonding wires **230** and **330** to the package substrate **100**. For example, the first bonding wires **230** may connect the first chip pads **220** of the first semiconductor chip **200** to first substrate pads **102a** of the package substrate **100**. The first bonding wires **230** may include first end portions coupled to the first chip pads **220**, second end portions coupled to the first substrate pads **102a**, and wire loops that connect the first end portions to the second end portions. When viewed in a plan view, the first substrate pads **102a** may be positioned in a direction opposite to the first direction **D1** of the first semiconductor chip **200**. The first substrate pads **102a** may be disposed

adjacent to the first lateral surface **200a** of the first semiconductor chip **200**. For example, the second bonding wires **330** may connect the second chip pads **320** of the second semiconductor chip **300** to second substrate pads **102b** of the package substrate **100**. The second bonding wires **330** may include third end portions coupled to the second chip pads **320**, fourth end portions coupled to the second substrate pads **102b**, and wire loops that connect the third end portions to the fourth end portions. When viewed in a plan view, the second substrate pads **102b** may be positioned in the first direction **D1** of the second semiconductor chip **300**. For example, the second substrate pads **102b** may be disposed adjacent to the third lateral surface **300a** of the second semiconductor chip **300**.

(27) The first bonding wires **230** may include third sub-wires **232** coupled to the third unit pads **222** and fourth sub-wires **234** coupled to the fourth unit pads **224**. For example, the first bonding wires **230** may not be connected to the chamfer region of the first semiconductor chip **200**. The second bonding wires **330** may include first sub-wires **332** coupled to the first unit pads **322** and second sub-wires **334** coupled to the second unit pads **324**. For example, the second bonding wires **330** may not be connected to the second region **R2** of the second semiconductor chip **300**.

(28) Bonding positions (or end portions of the first sub-wires **332**) of the first sub-wires **332** to the first unit pads **322** and bonding positions (or end portions of the second sub-wires **334**) of the second sub-wires **334** to the second unit pads **324** may depend on the first unit pads **322** and the second unit pads **324**. For example, the bonding positions (or end portions of the first sub-wires **332**) of the first sub-wires **332** to the first unit pads **322** may be arranged along the first line **L1** on the first region **R1**. The bonding positions (or end portions of the second sub-wires **334**) of the second sub-wires **334** to the second unit pads **324** may be arranged along the second line **L2** on the first region **R1** and may not be positioned on the second region **R2**.

(29) According to some example embodiments of the present inventive concepts, no pads may be provided adjacent to corners of the semiconductor chips **200** and **300**. Therefore, the semiconductor chips **200** and **300** may be applied with small pressure and torque in a wire bonding process of semiconductor package fabrication. Accordingly, a semiconductor package may not be damaged during fabrication thereof and may be provided with structural stability. This will be further discussed in detail below in describing a method of fabricating a semiconductor package.

(30) A molding layer **400** may be provided on the package substrate **100**. The molding layer **400** may cover the first semiconductor chip **200** and the second semiconductor chip **300**. The molding layer **400** may include a dielectric polymer material, such as an epoxy molding compound (EMC).

(31) For convenience of description, FIG. 4 illustrates a plan view of regions on a top surface of the second semiconductor chip, showing a semiconductor package according to some example embodiments of the present inventive concepts. For convenience of description, components the same as those of the example embodiments discussed with reference to FIGS. 1 to 3 are allocated the same reference numerals thereto, and a repetitive explanation thereof will be omitted or abridged below. The following description will focus on differences between the example embodiments of FIGS. 1 to 3 and other example embodiments described below.

(32) Referring to FIGS. 1 and 4, the first semiconductor chip **200** and the second semiconductor chip **300** may be provided on the package substrate **100**. The second chip pads **320** are adopted as a representative example to describe an arrangement and configuration of the first chip pads **220** of the first semiconductor chip **200** and an arrangement and configuration of the second chip pads **320** of the second semiconductor chip **300**.

(33) On the first region **R1**, the second chip pads **320** may include first unit pads **322** arranged along the first line **L1** and fifth unit pads **326** arranged along the second lines **L2**. The first unit pads **322** may be disposed adjacent to the third lateral surface **300a**. The first unit pads **322** may be arranged in the second direction **D2** so as to be parallel to the third lateral surface **300a**. The fifth unit pads **326** may be arranged along the second lines **L2** on the first region **R1**, and each of the fifth unit pads **326** may extend onto the second region **R2**. For example, the fifth unit pads **326** of

FIG. 4 may have their shapes obtained when the second unit pads 324 of FIG. 3 extend toward the third lateral surface 300a. The fifth unit pads 326 may have their linear shapes that extend toward the third lateral surface 300a from the first region R1 onto the second regions R2. When viewed in a plan view, the fifth unit pads 326 and the first unit pads 322 may have their lateral surfaces directed toward the third lateral surface 300a, and the lateral surfaces of the fifth unit pads 326 may be positioned on the same line on which the lateral surfaces of the first unit pads 322 are positioned. For example, a distance from the third lateral surface 300a to the fifth unit pads 326 may be substantially the same as that from the third lateral surface 300a to the first unit pads 322.

(34) The second semiconductor chip 300 may be wire-bonded to the package substrate 100. The second semiconductor chip 300 may be connected through the second bonding wires 330 to the package substrate 100.

(35) The second bonding wires 330 may include first sub-wires 332 coupled to the first unit pads 322 and fifth sub-wires 336 coupled to the fifth unit pads 326. Bonding positions (or end portions of the fifth sub-wires 336) of the fifth sub-wires 336 to the fifth unit pads 326 may be positioned on the first region R1. The closer to the fifth lateral surfaces 300c are disposed the bonding positions of the fifth sub-wires 336 to the fifth unit pads 326, the farther away from the third lateral surface 300a are disposed the bonding positions of the fifth sub-wires 336 to the fifth unit pads 326. For example, the second bonding wires 330 may not be bonded to the second region R2 of the second semiconductor chip 300. FIG. 4 shows that all of the fifth sub-wires 336 are coupled to the fifth unit pads 326 on the second region R2, but if necessary, the second sub-wires 334 may be correspondingly connected to the fifth unit pads 326 on the first region R1.

(36) The first semiconductor chip 200 may have the first chip pads 220 provided on the connection region. An arrangement and shape of the first chip pads 220 may be the same as or similar to an arrangement and shape of the second chip pads 320. The first chip pads 220 may be disposed adjacent to the first lateral surface 200a of the first semiconductor chip 200. For example, the first chip pads 220 may have third unit pads 222 and sixth unit pads 226. The third unit pads 222 may be arranged in the second direction D2 so as to be parallel to the first lateral surface 200a. The sixth unit pads 226 of FIG. 4 may have their shapes obtained when the fourth unit pads 224 of FIG. 3 extend toward the first lateral surface 200a. The sixth unit pads 226 may have their linear shapes that extend toward the first lateral surface 200a. When viewed in a plan view, the sixth unit pads 226 and the third unit pads 222 may have their lateral surfaces directed toward the first lateral surface 200a, and the lateral surfaces of the sixth unit pads 226 may be positioned on the same line on which the lateral surfaces of the third unit pads 222 are positioned.

(37) The first semiconductor chip 200 may be wire-bonded to the package substrate 100. The first semiconductor chip 200 may be connected through the first bonding wires 230 to the package substrate 100.

(38) The first bonding wires 230 may include third sub-wires 232 coupled to the third unit pads 222 and sixth sub-wires 236 coupled to the sixth unit pads 226. The closer the bonding positions of the sixth sub-wires 236 to the sixth unit pads 226 are to opposite lateral surfaces in the second direction D2 of the first semiconductor chip 200, the farther the bonding positions of the sixth sub-wires 236 to the sixth unit pads 226 are away from the first lateral surface 200a. For example, the first bonding wires 230 may not be bonded to the chamfer region of the first semiconductor chip 200. If necessary, the sixth sub-wires 236 may be connected to various positions on the sixth unit pads 226.

(39) For convenience of description, FIG. 5 illustrates a plan view of regions on a top surface of the second semiconductor chip, showing a semiconductor package according to some example embodiments of the present inventive concepts.

(40) Referring to FIGS. 1 and 5, the first semiconductor chip 200 and the second semiconductor chip 300 may be provided on the package substrate 100. The second chip pads 320 are adopted as a representative example to describe an arrangement and configuration of the first chip pads 220 of

the first semiconductor chip **200** and an arrangement and configuration of the second chip pads **320** of the second semiconductor chip **300**.

(41) The second chip pads **320** may include first unit pads **322** arranged along the first line **L1** on the first region **R1**, second unit pads **324** arranged along the second lines **L2** on the first region **R1**, and first dummy pads **328** disposed on the second regions **R2**. The first unit pads **322** may be disposed adjacent to the third lateral surface **300a**. The first unit pads **322** may be arranged in the second direction **D2** so as to be parallel to the third lateral surface **300a**. The closer to the fifth lateral surfaces **300c** the second unit pads **324** are disposed, the farther away from the third lateral surface **300a** the second unit pads **324** are disposed. The second chip pads **320** may not be provided on the second region **R2**. One or more first dummy pads **328** may be positioned in the first direction **D1** of one of the second unit pads **324**. At least one first dummy pad **328** may be disposed in the first direction **D1** of each of the second unit pads **324**. In this case, an increase in distance between the third lateral surface **300a** and the second unit pads **324** may induce an increase in the number of the first dummy pads **328** positioned in the first direction **D1** of the second unit pad **324**. The second unit pads **324** and the first dummy pads **328** may be arranged along the first direction **D1** and the second direction **D2**. A single pad group may be constituted by one second unit pad **324** and at least one first dummy pad **328** that is disposed in the first direction **D1** of the one second unit pad **324**. For example, the one second unit pad **324** and the at least one first dummy pad **328** that is arranged in the first direction **D1** of the one second unit pad **324** may be electrically connected to each other through an internal line **302** of the second semiconductor chip **300**. When viewed in a plan view, a lateral surface of the first dummy pad **328** most adjacent to the third lateral surface **300a** in one pad group may be positioned on the same line on which lateral surfaces of the first unit pads **322** directed toward the third lateral surface **300a**.

(42) The second semiconductor chip **300** may be wire-bonded to the package substrate **100**. The second semiconductor chip **300** may be connected through the second bonding wires **330** to the package substrate **100**.

(43) The second bonding wires **330** may include first sub-wires **332** coupled to the first unit pads **322** and second sub-wires **334** coupled to the second unit pads **324**. FIG. 5 shows that all of the second sub-wires **334** are coupled to the second unit pads **324**, but if necessary, in one pad group, each of the second sub-wires **334** may be connected to one of the first dummy pads **328** and the second unit pad **324** of the one pad group.

(44) The first semiconductor chip **200** may have the first chip pads **220** provided on the connection region. An arrangement and shape of the first chip pads **220** may be the same as or similar to an arrangement and shape of the second chip pads **320**. The first chip pads **220** may be disposed adjacent to the first lateral surface **200a** of the first semiconductor chip **200**. For example, the first chip pads **220** may have third unit pads **222**, fourth unit pads **224**, and second dummy pads **228**. The third unit pads **222** may be arranged in the second direction **D2** so as to be parallel to the first lateral surface **200a**. The fourth unit pads **224** may be disposed adjacent to corners of the first semiconductor chip **200**, and the closer the fourth unit pads **224** are to opposite lateral surfaces in the second direction **D2** of the first semiconductor chip **200**, the farther the fourth unit pads **224** are away from the first lateral surface **200a**. The second dummy pads **228** may be positioned in a direction opposite to the first direction **D1** of one of the fourth unit pads **224**. At least one second dummy pads **228** may be disposed in a direction opposite to the first direction **D1** of each of the fourth unit pads **224**. In this case, an increase in distance between the first lateral surface **200a** and the fourth unit pads **224** may induce an increase in the number of the second dummy pads **228** positioned in a direction opposite to the first direction **D1** of the fourth unit pad **224**. A single pad group may be constituted by one fourth unit pad **224** and at least one second dummy pad **228** that is arranged in a direction opposite to the first direction **D1** of the one fourth unit pad **224**. For example, the one fourth unit pad **224** and the at least one second dummy pad **228** that is arranged in a direction opposite to the first direction **D1** of the one fourth unit pad **224** may be electrically

connected to each other through an internal line **202** of the first semiconductor chip **200**. When viewed in a plan view, a lateral surface of the second dummy pad **228** most adjacent to the first lateral surface **200a** in one pad group may be positioned on the same line on which lateral surfaces of the third unit pads **222** directed toward the first lateral surface **200a**.

(45) The first semiconductor chip **200** may be wire-bonded to the package substrate **100**. The first semiconductor chip **200** may be connected through the first bonding wires **230** to the package substrate **100**.

(46) The first bonding wires **230** may include third sub-wires **232** coupled to the third unit pads **222** and fourth sub-wires **234** coupled to the fourth unit pads **224**. FIG. 5 shows that all of the fourth sub-wires **234** are coupled to the fourth unit pads **224**, but if necessary, in one pad group, each of the fourth sub-wires **234** may be connected to one of the second dummy pads **228** and the fourth unit pad **224** of the one pad group.

(47) FIG. 6 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts. FIGS. 7 to 9 illustrate plan views showing a semiconductor package according to some example embodiments of the present inventive concepts. FIG. 7 shows lowermost first and second semiconductor chips of a semiconductor package. FIG. 8 shows intermediate first and second semiconductor chips of a semiconductor package. FIG. 9 shows uppermost first and second semiconductor chips of a semiconductor package.

(48) Referring to FIG. 6, each of the first and second semiconductor chips **200** and **300** may be provided in plural. In the example embodiment of FIGS. 6 to 9, there are illustrated three first semiconductor chips **200-1**, **200-2**, and **200-3** and three second semiconductor chips **300-1**, **300-2**, and **300-3**, but the present inventive concepts are not limited thereto.

(49) The first semiconductor chips **200-1**, **200-2**, and **200-3** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may be stacked alternately with each other. The first semiconductor chips **200-1**, **200-2**, and **200-3** may be aligned in the third direction **D3**. For example, the first semiconductor chips **200-1**, **200-2**, and **200-3** may have lateral surfaces (each of which corresponds to the first lateral surface **200a** of FIG. 2) in a direction opposite to the first direction **D1**, and the lateral surfaces of the first semiconductor chips **200-1**, **200-2**, and **200-3** may be positioned on the same plane. The second semiconductor chips **300-1**, **300-2**, and **300-3** may be aligned in the third direction **D3**. For example, the second semiconductor chips **300-1**, **300-2**, and **300-3** may have lateral surfaces (each of which corresponds to the third lateral surface **300a** of FIG. 2) in the first direction **D1**, and the lateral surfaces of the second semiconductor chips **300-1**, **300-2**, and **300-3** may be positioned on the same plane. In this configuration, the first semiconductor chips **200-1**, **200-2**, and **200-3** may protrude the same distance in a direction opposite to the first direction **D1** from their adjacent second semiconductor chips **300-1**, **300-2**, and **300-3**, and the second semiconductor chips **300-1**, **300-2**, and **300-3** may protrude the same distance in the first direction **D1** from their adjacent first semiconductor chips **200-1**, **200-2**, and **200-3**. The first semiconductor chips **200-1**, **200-2**, and **200-3** may have their connection regions whose areas are the same as each other. The second semiconductor chips **300-1**, **300-2**, and **300-3** may have their connection regions **CR** whose areas are the same as each other.

(50) Each of the first semiconductor chips **200-1**, **200-2**, and **200-3** may be the same as or similar to the first semiconductor chip **200** discussed with reference to FIGS. 1 to 3. For example, the first semiconductor chips **200-1**, **200-2**, and **200-3** may have third unit pads **222** arranged along the first lateral surface **200a** and fourth unit pads **224** disposed adjacent to corners of the first semiconductor chips **200-1**, **200-2**, and **200-3**.

(51) Each of the second semiconductor chips **300-1**, **300-2**, and **300-3** may be the same as or similar to the second semiconductor chip **300** discussed with reference to FIGS. 1 to 3. For example, the second semiconductor chips **300-1**, **300-2**, and **300-3** may have first unit pads **322** arranged along the first line **L1** on the first region **R1** and second unit pads **324** arranged along the

second lines L2 on the first region R1.

(52) Referring to FIGS. 6 to 9, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in area of the second regions R2 on the connection region CR. For example, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce a reduction in length of the first line L1 and an increase in length of the second lines L2. Alternatively, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in distance between the second lines L2 and corners of the second semiconductor chips **300-1**, **300-2**, and **300-3**. An increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce a reduction in the number of the first unit pads **322** adjacent to the third lateral surface **300a** and an increase in the number of the second unit pads **324** far away from the third lateral surface **300a**. An increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in distance between the third lateral surface **300a** and the second unit pads **324**. For example, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in distance between the second chip pads **320** and corners of the second semiconductor chips **300-1**, **300-2**, and **300-3**.

(53) The second semiconductor chips **300-1**, **300-2**, and **300-3** may be wire-bonded to the package substrate **100**. The second semiconductor chips **300-1**, **300-2**, and **300-3** may be connected through the second bonding wires **330** to the package substrate **100**.

(54) The second bonding wires **330** may include first sub-wires **332** coupled to the first unit pads **322** and second sub-wires **334** coupled to the second unit pads **324**. Based on positions of the second unit pads **324**, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in distance between the third lateral surface **300a** and bonding positions of the second sub-wires **334** to the second unit pads **324**.

(55) An arrangement and shape of the first semiconductor chips **200-1**, **200-2**, and **200-3** may be the same as or similar to an arrangement and shape of the second semiconductor chips **300-1**, **300-2**, and **300-3**. For example, an increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce an increase in area of chamfer regions on corners of the first semiconductor chips **200-1**, **200-2**, and **200-3**. An increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce a reduction in the number of the third unit pads **222** adjacent to the first lateral surface **200a** and an increase in the number of the fourth unit pads **224** far away from the first lateral surface **200a**. An increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce an increase in distance between the first lateral surface **200a** and the fourth unit pads **224**. For example, an increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce an increase in distance between the first chip pads **220** and corners of the first semiconductor chips **200-1**, **200-2**, and **200-3**.

(56) The first semiconductor chips **200-1**, **200-2**, and **200-3** may be wire-bonded to the package substrate **100**. The first semiconductor chips **200-1**, **200-2**, and **200-3** may be connected through the first bonding wires **230** to the package substrate **100**.

(57) The first bonding wires **230** may include third sub-wires **232** coupled to the third unit pads **222** and fourth sub-wires **234** coupled to the fourth unit pads **224**. Based on positions of the fourth unit pads **224**, an increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce an increase in distance between the first lateral surface **200a** and bonding positions of the fourth sub-wires **234** to the fourth unit pads **224**.

(58) In general, an increase in distance between semiconductor chips and the package substrate **100** may induce an increase in pressure and torque applied to the semiconductor chips in a wire bonding process of semiconductor package fabrication.

(59) According to some example embodiments of the present inventive concepts, an increase in distance between the package substrate **100** and the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** may induce an increase in area of the second regions R2 or sections where no pads are provided on corners of the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3**. Therefore, in a wire bonding process of semiconductor package fabrication, there may be a reduction in pressure and torque applied to high-level placed ones of the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3**. Accordingly, a semiconductor package may not be damaged during fabrication thereof and may be provided with structural stability.

(60) FIGS. **10** to **12** illustrate plan views showing a semiconductor package according to some example embodiments of the present inventive concepts. FIG. **10** shows lowermost first and second semiconductor chips of a semiconductor package. FIG. **11** shows intermediate first and second semiconductor chips of a semiconductor package. FIG. **12** shows uppermost first and second semiconductor chips of a semiconductor package.

(61) Referring to FIG. **6**, each of the first semiconductor chips **200-1**, **200-2**, and **200-3** may be the same as or similar to the first semiconductor chip **200** discussed with reference to FIGS. **1** and **4**. For example, the first semiconductor chips **200-1**, **200-2**, and **200-3** may have third unit pads **222** arranged along the first lateral surface **200a** and sixth unit pads **226** disposed adjacent to corners of the first semiconductor chips **200-1**, **200-2**, and **200-3**. Each of the sixth unit pads **226** may extend toward the first lateral surface **200a**. The first semiconductor chips **200-1**, **200-2**, and **200-3** may have the same configuration and arrangement of the first chip pads **220**. For example, the first semiconductor chips **200-1**, **200-2**, and **200-3** may be the same chips.

(62) Each of the second semiconductor chips **300-1**, **300-2**, and **300-3** may be the same as or similar to the second semiconductor chip **300** discussed with reference to FIGS. **1** and **4**. For example, the second semiconductor chips **300-1**, **300-2**, and **300-3** may have first unit pads **322** arranged along the first line L1 on the first region R1 and fifth unit pads **326** arranged along the second lines L2 on the first region RE. The fifth unit pads **326** may be arranged along the second lines L2 on the first region R1, and each of the fifth unit pads **326** may extend onto the second region R2. The second semiconductor chips **300-1**, **300-2**, and **300-3** may have the same configuration and arrangement of the second chip pads **320**. For example, the second semiconductor chips **300-1**, **300-2**, and **300-3** may be the same chips.

(63) Referring to FIGS. **6** and **10** to **12**, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in area of the second regions R2 on the connection region CR. In this description, regions on semiconductor chips may indicate arbitrary regions, and the fact that the semiconductor chips have different sized regions may not mean that the semiconductor chips have different shapes and configurations. An increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce a reduction in length of the first line L1 and an increase in length of the second lines L2. Alternatively, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in distance between the second lines L2 and corners of the second semiconductor chips **300-1**, **300-2**, and **300-3**.

(64) The second semiconductor chips **300-1**, **300-2**, and **300-3** may be wire-bonded to the package substrate **100**. The second semiconductor chips **300-1**, **300-2**, and **300-3** may be connected through the second bonding wires **330** to the package substrate **100**.

(65) The second bonding wires **330** may include first sub-wires **332** coupled to the first unit pads **322** and second sub-wires **334** coupled to the fifth unit pads **326**. In this case, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in distance between the third lateral surface **300a** and bonding positions of the second sub-wires **334** to the fifth unit pads **326**. For example, the bonding positions of the second sub-wires **334** to the fifth unit pads **326** may be positioned along the second lines L2.

(66) An arrangement and shape of the first semiconductor chips **200-1**, **200-2**, and **200-3** may be the same as or similar to an arrangement and shape of the second semiconductor chips **300-1**, **300-2**, and **300-3**. For example, an increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce an increase in area of chamfer regions on corners of the first semiconductor chips **200-1**, **200-2**, and **200-3**.

(67) The first semiconductor chips **200-1**, **200-2**, and **200-3** may be wire-bonded to the package substrate **100**. The first semiconductor chips **200-1**, **200-2**, and **200-3** may be connected through the first bonding wires **230** to the package substrate **100**.

(68) The first bonding wires **230** may include third sub-wires **232** coupled to the third unit pads **222** and fourth sub-wires **234** coupled to the sixth unit pads **226**. Based on positions of the sixth unit pads **226**, an increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce an increase in distance between the first lateral surface **200a** and bonding positions of the fourth sub-wires **234** to the sixth unit pads **226**.

(69) According to some example embodiments of the present inventive concepts, an increase in distance between the package substrate **100** and the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** may induce an increase in distance of bonding positions of the bonding wires **230** and **330** between corners of the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3**. In addition, as the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** have large area pads **226** and **326** adjacent to corners thereof, it may be possible to stack the same semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** and simultaneously to allow the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** to have different bonding positions. Therefore, it may be possible to reduce pressure and torque applied to high-level placed ones of the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** while stacking the same semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3**. Accordingly, a semiconductor package may not be damaged during fabrication thereof and may be provided with structural stability.

(70) FIGS. **13** to **15** illustrate plan views showing a semiconductor package according to some example embodiments of the present inventive concepts. FIG. **13** shows lowermost first and second semiconductor chips of a semiconductor package. FIG. **14** shows intermediate first and second semiconductor chips of a semiconductor package. FIG. **15** shows uppermost first and second semiconductor chips of a semiconductor package.

(71) Referring to FIG. **6**, each of the first semiconductor chips **200-1**, **200-2**, and **200-3** may be the same as or similar to the first semiconductor chip **200** discussed with reference to FIGS. **1** and **5**. For example, the first semiconductor chips **200-1**, **200-2**, and **200-3** may have third unit pads **222** arranged along the first lateral surface **200a**, fourth unit pads **224** disposed adjacent to corners of the first semiconductor chips **200-1**, **200-2**, and **200-3**, and second dummy pads **228** provided in a direction opposite to the first direction **D1** of the fourth unit pads **224**. A single pad group may be constituted by one fourth unit pad **224** and at least one second dummy pad **228** that is arranged in a direction opposite to the first direction **D1** of the one fourth unit pad **224**. The first semiconductor chips **200-1**, **200-2**, and **200-3** may have the same configuration and arrangement of the first chip pads **220**. For example, the first semiconductor chips **200-1**, **200-2**, and **200-3** may be the same chips.

(72) Each of the second semiconductor chips **300-1**, **300-2**, and **300-3** may be the same as or similar to the second semiconductor chip **300** discussed with reference to FIGS. **1** and **5**. For example, the second semiconductor chips **300-1**, **300-2**, and **300-3** may have first unit pads **322** arranged along the first line **L1** on the first region **R1**, second unit pads **324** arranged along the second lines **L2** on the first region **R1**, and first dummy pads **328** disposed on the second regions **R2**. A single pad group may be constituted by one second unit pad **324** and at least one first dummy pad **328** that is disposed in the first direction **D1** of the one second unit pad **324**. The second semiconductor chips **300-1**, **300-2**, and **300-3** may have the same configuration and arrangement of

the second chip pads **320**. For example, the second semiconductor chips **300-1**, **300-2**, and **300-3** may be the same chips.

(73) Referring to FIGS. **6** and **13** to **15**, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in area of the second regions **R2** on the connection region **CR**. An increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce a reduction in length of the first line **L1** and an increase in length of the second lines **L2**. Alternatively, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in distance between the second lines **L2** and corners of the second semiconductor chips **300-1**, **300-2**, and **300-3**.

(74) The second semiconductor chips **300-1**, **300-2**, and **300-3** may be wire-bonded to the package substrate **100**. The second semiconductor chips **300-1**, **300-2**, and **300-3** may be connected through the second bonding wires **330** to the package substrate **100**.

(75) The second bonding wires **330** may include first sub-wires **332** coupled to the first unit pads **322**, and may also include second sub-wires **334** coupled to one of the first dummy pads **328** and the second unit pad **324** of the one pad group. In this case, an increase in distance between the package substrate **100** and the second semiconductor chips **300-1**, **300-2**, and **300-3** may induce an increase in distance between the third lateral surface **300a** and bonding positions of the second sub-wires **334** to the first dummy pads **328** and the second unit pads **324** of the pad groups. For example, the bonding positions of the second sub-wires **334** to the first dummy pads **328** and the second unit pads **324** of the pad group may be positioned along the second lines **L2**.

(76) An arrangement and shape of the first semiconductor chips **200-1**, **200-2**, and **200-3** may be the same as or similar to an arrangement and shape of the second semiconductor chips **300-1**, **300-2**, and **300-3**. For example, an increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce an increase in area of chamfer regions on corners of the first semiconductor chips **200-1**, **200-2**, and **200-3**.

(77) The first semiconductor chips **200-1**, **200-2**, and **200-3** may be wire-bonded to the package substrate **100**. The first semiconductor chips **200-1**, **200-2**, and **200-3** may be connected through the first bonding wires **230** to the package substrate **100**.

(78) The first bonding wires **230** may include third sub-wires **232** coupled to the third unit pads **222**, and may also include fourth sub-wires **234** coupled to one of the second dummy pads **228** and the fourth unit pad **224** of the one pad group. In this case, an increase in distance between the package substrate **100** and the first semiconductor chips **200-1**, **200-2**, and **200-3** may induce an increase in distance between the first lateral surface **200a** and bonding positions of the fourth sub-wires **234** to the second dummy pads **228** and the fourth unit pads **224** of the pad groups.

(79) According to some example embodiments of the present inventive concepts, as the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** have the pads **224** and **324** and the dummy pads **228** and **328** connected to the pads **224** and **324**, it may be possible to stack the same semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** and simultaneously to allow the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** to have different bonding positions. Therefore, it may be possible to reduce pressure and torque applied to high-level placed ones of the semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3** while stacking the same semiconductor chips **200-1**, **300-1**, **200-2**, **300-2**, **200-3**, and **300-3**. Accordingly, a semiconductor package may not be damaged during fabrication thereof and may be provided with structural stability.

(80) FIGS. **16** and **17** illustrate cross-sectional views showing a semiconductor package according to some example embodiments of the present inventive concepts. FIGS. **18** to **20** illustrate plan views showing a semiconductor package according to some example embodiments of the present inventive concepts.

(81) Referring to FIGS. **16** and **17**, each of the first and second semiconductor chips **200** and **300**

may be provided in plural. The first semiconductor chips **200-4**, **200-5**, and **200-6** and the second semiconductor chips **300-4**, **300-5**, and **300-6** may be stacked alternately with each other. In this case, an increase in distance between the package substrate **100** and the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce a reduction in protruding length in a direction opposite to the first direction **D1** of the first semiconductor chips **200-4**, **200-5**, and **200-6** from overlying second semiconductor chips **300-4**, **300-5**, and **300-6**. An increase in distance between the package substrate **100** and the second semiconductor chips **300-4**, **300-5**, and **300-6** may induce a reduction in protruding length in the first direction **D1** of the second semiconductor chips **300-4**, **300-5**, and **300-6** from underlying first semiconductor chips **200-4**, **200-5**, and **200-6**. For example, the first semiconductor chips **200-4**, **200-5**, and **200-6** may be stacked in a cascade shape, and the second semiconductor chips **300-4**, **300-5**, and **300-6** may be stacked in a cascade shape. An increase in distance between the package substrate **100** and the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce a reduction in area of the connection regions of the first semiconductor chips **200-4**, **200-5**, and **200-6**. An increase in distance between the package substrate **100** and the second semiconductor chips **300-4**, **300-5**, and **300-6** may induce a reduction in area of the connection regions **CR** of the second semiconductor chips **300-4**, **300-5**, and **300-6**. The present inventive concepts, however, are not limited thereto. The example embodiment of FIGS. **16** to **20** is provided to explain a semiconductor package in which semiconductor chips have different protruding lengths. In the example embodiment of FIGS. **16** to **20**, the first and second semiconductor chips **200-4** and **300-4** may have larger protruding lengths, the first and second semiconductor chips **200-6** and **300-6** may have smaller protruding lengths, and the first and second semiconductor chips **200-5** and **300-5** may have medium protruding lengths.

(82) Referring to FIGS. **16** to **20**, an increase in protruding length of the second semiconductor chips **300-4**, **300-5**, and **300-6** from the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce an increase in area of the second regions **R2** on the connection region **CR**. For example, an increase in protruding length of the second semiconductor chips **300-4**, **300-5**, and **300-6** from the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce a reduction in length of the first line **L1** and an increase in length of the second lines **L2**. Alternatively, an increase in protruding length of the second semiconductor chips **300-4**, **300-5**, and **300-6** from the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce an increase in distance between the second lines **L2** and corners of the second semiconductor chips **300-4**, **300-5**, and **300-6**. An increase in protruding length of the second semiconductor chips **300-4**, **300-5**, and **300-6** from the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce a reduction in the number of the first unit pads **322** adjacent to the third lateral surface **300a** and an increase in the number of the second unit pads **324** far away from the third lateral surface **300a**. An increase in protruding length of the second semiconductor chips **300-4**, **300-5**, and **300-6** from the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce an increase in distance between the third lateral surface **300a** and the second unit pads **324**. For example, an increase in protruding length of the second semiconductor chips **300-4**, **300-5**, and **300-6** from the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce an increase in distance between the second chip pads **320** and corners of the second semiconductor chips **300-4**, **300-5**, and **300-6**.

(83) The second semiconductor chips **300-4**, **300-5**, and **300-6** may be wire-bonded to the package substrate **100**. As shown in FIG. **16**, each of the second semiconductor chips **300-4**, **300-5**, and **300-6** may be connected through the second bonding wires **330** to the package substrate **100**. For example, each of the second bonding wires **330** may connect one of the second chip pads **320** of the second semiconductor chips **300-4**, **300-5**, and **300-6** to one of the second substrate pads **102b** of the package substrate **100**. Alternatively, as shown in FIG. **17**, the package substrate **100** and the second semiconductor chips **300-4**, **300-5**, and **300-6** may be connected to each other through the second bonding wires **330**. For example, each of the second bonding wires **330** may connect to each other the second chip pads **320** of the second semiconductor chips **300-4**, **300-5**, and **300-6**, or

may connect one of the second chip pads **320** to one of the second substrate pads **102b** of the package substrate **100**.

(84) The second bonding wires **330** may include first sub-wires **332** coupled to the first unit pads **322** and second sub-wires **334** coupled to the second unit pads **324**. Based on positions of the second unit pads **324**, an increase in protruding length of the second semiconductor chips **300-4**, **300-5**, and **300-6** from the first semiconductor chips **200-4**, **200-5**, and **200-6** may induce an increase in distance between the third lateral surface **300a** and bonding positions of the second sub-wires **334** to the second unit pads **324**.

(85) An arrangement and shape of the first semiconductor chips **200-4**, **200-5**, and **200-6** may be the same as or similar to an arrangement and shape of the second semiconductor chips **300-4**, **300-5**, and **300-6**. For example, an increase in protruding length of the first semiconductor chips **200-4**, **200-5**, and **200-6** from the second semiconductor chips **300-4**, **300-5**, and **300-6** may induce an increase in area of chamfer regions on corners of the first semiconductor chips **200-4**, **200-5**, and **200-6**. An increase in protruding length of the first semiconductor chips **200-4**, **200-5**, and **200-6** from the second semiconductor chips **300-4**, **300-5**, and **300-6** may induce a reduction in the number of the third unit pads **222** adjacent to the first lateral surface **200a** and an increase in the number of the fourth unit pads **224** far away from the first lateral surface **200a**. An increase in protruding length of the first semiconductor chips **200-4**, **200-5**, and **200-6** from the second semiconductor chips **300-4**, **300-5**, and **300-6** may induce an increase in distance between the first lateral surface **200a** and the fourth unit pads **224**. For example, an increase in protruding length of the first semiconductor chips **200-4**, **200-5**, and **200-6** from the second semiconductor chips **300-4**, **300-5**, and **300-6** may induce an increase in distance between the first chip pads **220** and corners of the first semiconductor chips **200-4**, **200-5**, and **200-6**.

(86) The first semiconductor chips **200-4**, **200-5**, and **200-6** may be wire-bonded to the package substrate **100**. As shown in FIG. 16, each of the first semiconductor chips **200-4**, **200-5**, and **200-6** may be connected through the first bonding wires **230** to the package substrate **100**. For example, each of the first bonding wires **230** may connect one of the first chip pads **220** of the first semiconductor chips **200-4**, **200-5**, and **200-6** to one of the first substrate pads **102a** of the package substrate **100**. Alternatively, as shown in FIG. 17, the package substrate **100** and the first semiconductor chips **200-4**, **200-5**, and **200-6** may be connected to each other through the first bonding wires **230**. For example, each of the first bonding wires **230** may connect to each other the first chip pads **220** of the first semiconductor chips **200-4**, **200-5**, and **200-6**, or may connect one of the first chip pads **220** to one of the first substrate pads **102a** of the package substrate **100**.

(87) The first bonding wires **230** may include third sub-wires **232** coupled to the third unit pads **222** and fourth sub-wires **234** coupled to the fourth unit pads **224**. Based on positions of the fourth unit pads **224**, an increase in protruding length of the first semiconductor chips **200-4**, **200-5**, and **200-6** from the second semiconductor chips **300-4**, **300-5**, and **300-6** may induce an increase in distance between the first lateral surface **200a** and bonding positions of the fourth sub-wires **234** to the fourth unit pads **224**.

(88) In general, an increase in protruding length of semiconductor chips from an adjacent other semiconductor chip may induce an increase in pressure and torque applied to the semiconductor chips in a wire bonding process of semiconductor package fabrication.

(89) According to some example embodiments of the present inventive concepts, an increase in protruding length of the semiconductor chips **200-4**, **300-4**, **200-5**, **300-5**, **200-6**, and **300-6** from an adjacent other semiconductor chip may induce an increase in area of the second regions **R2** or sections where no pads are provided on corners of the semiconductor chips **200-4**, **300-4**, **200-5**, **300-5**, **200-6**, and **300-6**. Therefore, in a wire bonding process of semiconductor package fabrication, there may be a reduction in pressure and torque applied to the semiconductor chips **200-4**, **300-4**, **200-5**, **300-5**, **200-6**, and **300-6** having large protruding lengths from an adjacent other semiconductor chip. Accordingly, a semiconductor package may not be damaged during

fabrication thereof and may be provided with structural stability.

(90) FIGS. **21A** to **25A** and **23C** illustrate cross-sectional views showing a method of fabricating a semiconductor package according to some example embodiments of the present inventive concepts. FIGS. **21B** to **25B** illustrate plan views showing a method of fabricating a semiconductor package according to some example embodiments of the present inventive concepts. FIGS. **21A** to **25A** correspond to cross-sectional views taken along line I-I' of FIGS. **21B** to **25B**, respectively. FIG. **23C** corresponds to a cross-sectional view taken along line II-II' of FIG. **23B**.

(91) Referring to FIGS. **21A** and **21B**, a package substrate **100** may be provided. The package substrate **100** may include first and second substrate pads **102a** and **102b** provided on a top surface of the package substrate **100**.

(92) A first semiconductor chip **200** may be attached to the package substrate **100**. The first semiconductor chip **200** may be attached in a face-up state to the package substrate **100**. The first semiconductor chip **200** may have a top surface **200u** or an active surface. A first adhesive layer **210** may be used to attach the first semiconductor chip **200** to the package substrate **100**. For example, the first adhesive layer **210** may be provided on a bottom surface of the first semiconductor chip **200**, and may adhere the first semiconductor chip **200** to the top surface of the package substrate **100**.

(93) The first semiconductor chip **200** may be wire-bonded to the package substrate **100**. The first semiconductor chip **200** may be connected through first bonding wires **230** to the package substrate **100**. For example, the first bonding wires **230** may connect first chip pads **220** of the first semiconductor chip **200** to the first substrate pads **102a** of the package substrate **100**.

(94) Referring to FIGS. **22A** and **22B**, a second semiconductor chip **300** may be attached to the first semiconductor chip **200**. The second semiconductor chip **300** may be attached in a face-up state to the first semiconductor chip **200**. The second semiconductor chip **300** may have a top surface **300u** or an active surface. A second adhesive layer **310** may be used to attach the second semiconductor chip **300** to the first semiconductor chip **200**. For example, the second adhesive layer **310** may be provided on a bottom surface of the second semiconductor chip **300**, and may adhere the second semiconductor chip **300** to a top surface of the first semiconductor chip **200**. The second semiconductor chip **300** may be shifted from the first semiconductor chip **200** in a direction opposite to the first direction **D1**. For example, a connection region **CR** of the second semiconductor chip **300** may not overlap the first semiconductor chip **200**, and an empty space may be provided between the package substrate **100** and the connection region **CR** of the second semiconductor chip **300**. For example, no component may be provided to support the connection region **CR** of the second semiconductor chip **300**.

(95) Second chip pads **320** may include first unit pads **322** arranged along a first line **L1** on a first region **R1** and second unit pads **324** arranged along second lines **L2** on the first region **RE**. The first unit pads **322** may be disposed adjacent to a third lateral surface **300a**. The first unit pads **322** may be arranged in the second direction **D2** so as to be parallel to the third lateral surface **300a**. The closer to fifth lateral surfaces **300c** the second unit pads **324** are disposed, the farther away from the third lateral surface **300a** the second unit pads **324** are disposed. The second chip pads **320** may not be provided on a second region **R2**.

(96) Referring to FIGS. **23A**, **23B**, and **23C**, the second semiconductor chip **300** may be wire-bonded to the package substrate **100**. The second semiconductor chip **300** may be connected through second bonding wires **330** to the package substrate **100**. FIGS. **23A**, **23B**, and **23C** explain that the second unit pads **324** of the second chip pads **320** of the second semiconductor chip **300** are connected to the package substrate **100** in a wire bonding process of the second semiconductor chip **300**.

(97) A bonding machine **MA** may descend to come into contact with a top surface of one of the second unit pads **324**. The bonding machine **MA** may spray a solder material to form a solder part **BM** on the top surface of the second unit pad **324**.

(98) Afterwards, the bonding machine MA may move onto one of the second substrate pads **102b** of the package substrate **100**, while spraying the solder material. In this step, the solder material sprayed from the bonding machine MA may form a second sub-wire **334** that connects one of the second unit pads **324** to one of the second substrate pads **102b**.

(99) Referring to FIGS. **24A** and **24B**, the second semiconductor chip **300** may be wire-bonded to the package substrate **100**. FIGS. **24A** and **24B** explain that the first unit pads **322** of the second chip pads **320** of the second semiconductor chip **300** are connected to the package substrate **100** in a wire bonding process of the second semiconductor chip **300**.

(100) The bonding machine MA may descend to come into contact with a top surface of one of the first unit pads **322**. The bonding machine MA may spray a solder material to form a solder part BM on the top surface of the first unit pad **322**. In this step, a pressure applied by the bonding machine MA to the first unit pad **322** may be the same as a pressure applied by the bonding machine MA to the second unit pads **324** in a wire bonding process performed on the second unit pads **324**. The present inventive concepts, however, are not limited thereto.

(101) After that, the bonding machine MA may move onto one of the second substrate pads **102b** of the package substrate **100** while spraying the solder material. In this step, the solder material sprayed from the bonding machine MA may form a first sub-wire **332** that connects one of the first unit pads **322** to one of the second substrate pads **102b**.

(102) In general, during a bonding process for wire bonding, a bonding machine may bring into contact with pads, and in this case, may provide the pads with vertical pressure. When a certain pressure is applied to one point on a semiconductor chip, the largest torque may be exerted on a corner of the semiconductor chip, and the corner of the semiconductor chip may be most vulnerable to fracture. For example, as shown in FIGS. **25A** and **25B**, when pads **320'** of the second semiconductor chip **300** are disposed adjacent to a corner of the second semiconductor chip **300**, the bonding machine MA may compress the pad **320'** adjacent to the corner of the second semiconductor chip **300**, and in this case, the bonding machine MA may cause fracture of the second semiconductor chip **300**.

(103) According to some example embodiments of the present inventive concepts, the connection region CR of the second semiconductor chip **300** may not be supported by the first semiconductor chip **200**, and an empty space may be provided between the package substrate **100** and the connection region CR of the second semiconductor chip **300**. No pad may be provided on the second region R2 in the vicinity of a corner of the second semiconductor chip **300**, and therefore, a pressure exerted by the bonding machine MA may not be applied to the second region R2 of the second semiconductor chip **300**. Accordingly, it may be possible to cause less fracture of semiconductor chips in semiconductor fabrication process and to provide a semiconductor fabrication method with less occurrence of failure. In addition, because no pads are provided on the second region R2 of the second semiconductor chip **300**, it may be possible to allow the bonding machine MA to rapidly form bonding parts on pads at high pressure and to provide a semiconductor fabrication method with high speed.

(104) In a semiconductor package according to some example embodiments of the present inventive concepts, small pressure and torque may be applied to semiconductor chips in a wire bonding process of semiconductor package fabrication. In particular, it may be possible to reduce pressure and torque applied to semiconductor chips disposed at high levels while stacking the same semiconductor chips, and to decrease pressure and torque exerted on semiconductor chips having large protruding lengths from an adjacent other semiconductor chip. It may thus be possible to provide a semiconductor fabrication method with less occurrence of failure and to provide a semiconductor package with increased structural stability.

(105) It will be understood that elements and/or properties thereof described herein that are described with a numerical value include a tolerance of $\pm 10\%$ around the stated numerical value.

(106) Moreover, it may be possible to allow a bonding machine to rapidly form bonding parts on

pads at high pressure and to provide a semiconductor fabrication method with high speed.
(107) Although the present inventive concepts have been described in connection with some example embodiments of the present inventive concepts illustrated in the accompanying drawings, it will be understood by one of ordinary skill in the art that variations in form and detail may be made therein without departing from the spirit and essential feature of the present inventive concepts. The above disclosed example embodiments should thus be considered illustrative and not restrictive.

Claims

1. A semiconductor package, comprising: a substrate; a plurality of semiconductor chips stacked on the substrate and each of the plurality of semiconductor chips including a plurality of first pads and a plurality of second pads on a top surface thereof; and a plurality of bonding wires that connect the plurality of first pads and the plurality of second pads of the plurality of semiconductor chips to the substrate, wherein the plurality of semiconductor chips each protrude relative to adjacent ones of the plurality of semiconductor chips alternately in a first direction and a direction opposite to the first direction, and protruding distances of the plurality of semiconductor chips being same as each other, wherein the plurality of semiconductor chips each have a first lateral surface spaced apart from another of the plurality of semiconductor chips in overlying or underlying contact therewith, wherein the top surface of each of the plurality of semiconductor chips is provided thereon with, a first arrangement line that extends along the first lateral surface; and a plurality of second arrangement lines that extend from opposite ends of the first arrangement line, wherein as the second arrangement lines increase in distance from the first arrangement line, the second arrangement lines increase in distance from the first lateral surface, wherein the plurality of first pads are arranged along the first arrangement line, and wherein the plurality of second pads are arranged along the second arrangement lines.
2. The semiconductor package of claim 1, wherein the first arrangement line is a straight line parallel to the first lateral surface, and the second arrangement lines are straight lines that are inclined at 45 degrees relative to the first lateral surface.
3. The semiconductor package of claim 1, wherein an increase in distance between the substrate and the plurality of semiconductor chips includes a reduction in length of the first arrangement line and an increase in length of each of the second arrangement lines.
4. The semiconductor package of claim 1, wherein the top surface of each of the semiconductor chips has a plurality of chamfer regions defined by the second arrangement lines, the first lateral surface, and a plurality of second lateral surfaces in contact with the first lateral surface, and no pad is provided on the chamfer regions.
5. The semiconductor package of claim 1, wherein each of the plurality of second pads of each of the plurality of semiconductor chips has a linear shape that extends toward the first lateral surface.
6. The semiconductor package of claim 5, wherein as a distance between the substrate and the plurality of semiconductor chips increases, a distance between the first lateral surface and bonding positions of the bonding wires to the plurality of second pads increases.
7. The semiconductor package of claim 1, wherein each of the plurality of semiconductor chips further includes a plurality of dummy pads arranged in a second direction toward the first lateral surface and away from the plurality of second pads, and a single pad group is constituted by one of the plurality of second pads and at least one of the plurality of dummy pads, the at least one dummy pad being positioned in the second direction of the one second pad.
8. The semiconductor package of claim 7, wherein the bonding wire is bonded to one selected from the plurality of dummy pads and the one second pad of the pad group, and wherein as a distance between the substrate and the plurality of semiconductor chips increases, a distance between the first lateral surface and a bonding position of the bonding wire to the one dummy pad and the one

second pad of the pad group increases.

9. The semiconductor package of claim 1, wherein the plurality of semiconductor chips include a plurality of first semiconductor chips and a plurality of second semiconductor chips that are alternately stacked with each other on the substrate, wherein the first lateral surfaces of the plurality of first semiconductor chips are on a same first plane, and the first lateral surfaces of the plurality of second semiconductor chips are on a same second plane.

10. The semiconductor package of claim 9, wherein the plurality of semiconductor chips are the same, and planar shapes of the plurality of first semiconductor chips are symmetric in the first direction to planar shapes of the plurality of second semiconductor chips.
